

Steeling Against Time : A Case of the Korean Steel Industry

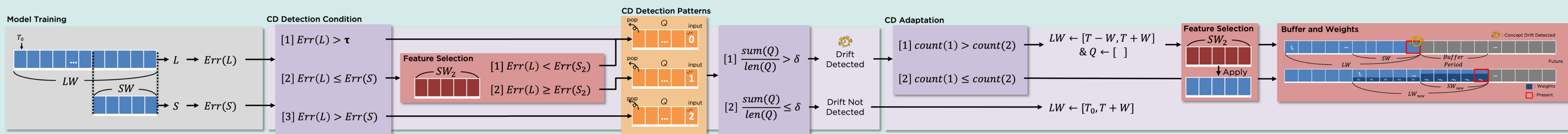
Jeong-Uk Seo, SeongHyun Seo, Dong-Joon Lim



Introduction

Concept drift (CD) refers to a situation in which the statistical properties of the target variable, which the model aims to predict, unpredictably change over time. Methods for **CD detection (CDD)** and **CD adaptation (CDA)** provide insights into scenarios where model performance declines due to these unpredictable changes. We propose an **adaptive framework** that facilitates CDD and CDA. Additionally, during a '**buffer period**', various '**weights**' are applied to update the model. An empirical case study from **the Korean steel industry** highlights the necessity of **autonomous retraining strategies**.

Adaptive Model Update Framework



Model Update Strategy - Managements of Model and Data

Model Update Strategy

- It involves updating the model non-periodically based on CDD.
- It can be divided into two main aspects:

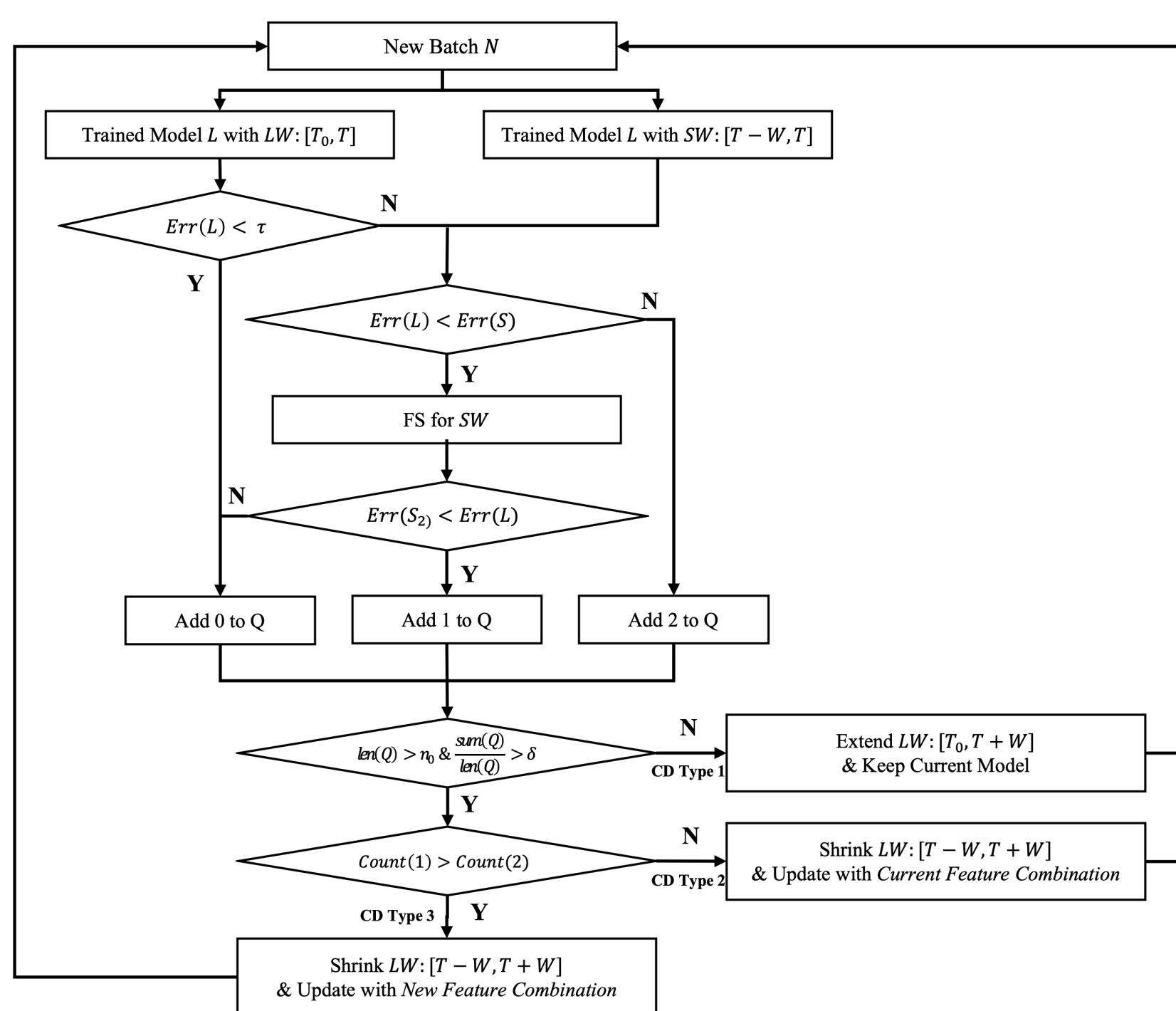
Model Management and Data Management.

Managements	Explanation
Model	- The systematic approach for overseeing the entire lifecycle of ML models from their development through deployment and ongoing maintenance. - Involves optimizing the model itself, such as using Feature Selection(FS). - Helps model to enhance performance and adapting it to changing data patterns.
Data	- Concerns how to structure the data for retraining the ML model. - Essential to make optimal use of the most recent data where changes have occurred. - Essential to focus on the quantity and importance of data in model update. - Helps to adjust the sensitivity during training.

Model Management with CD Detection and Adaptation

Double Window based Concept Drift Detection Method

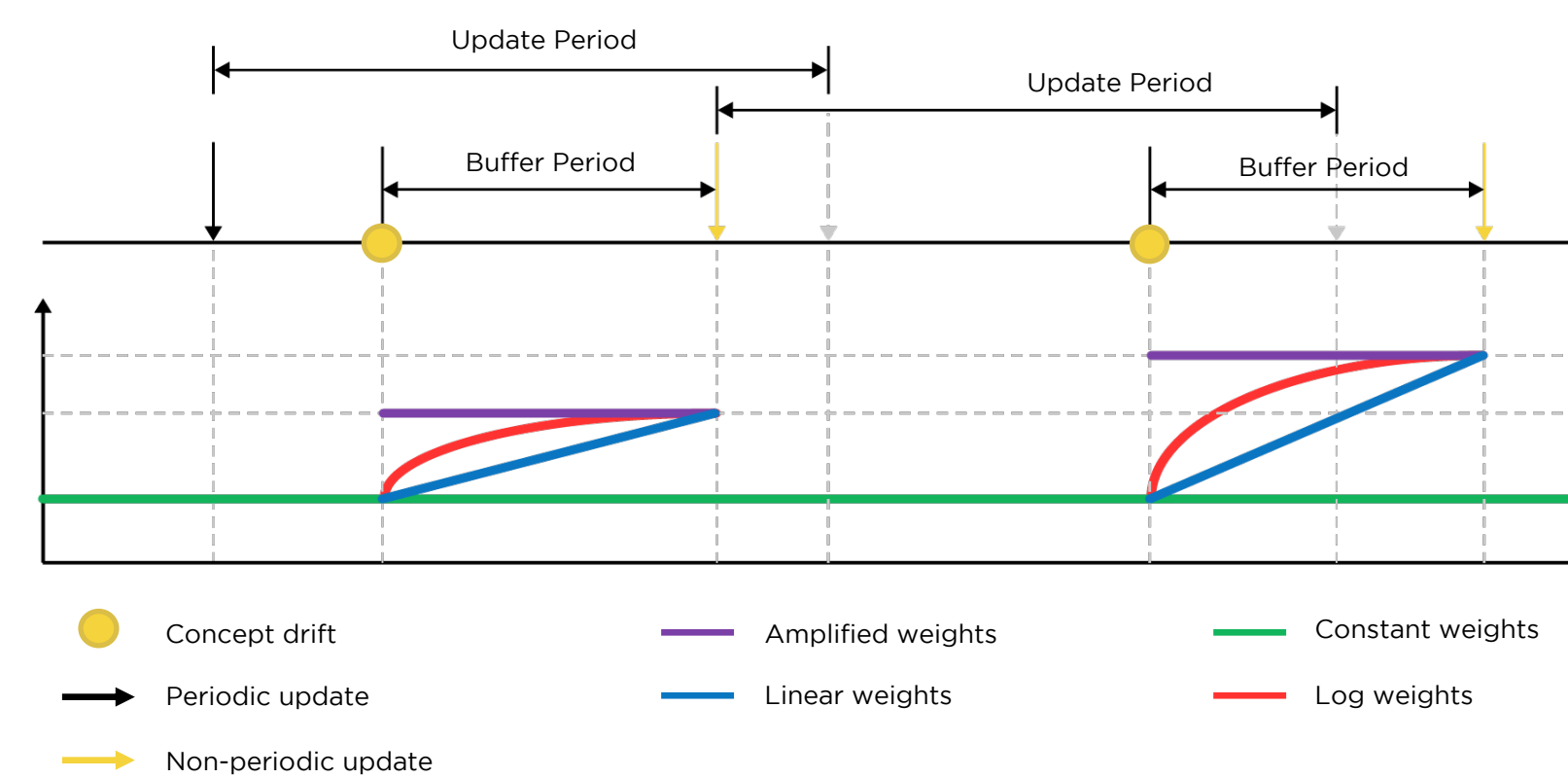
- Three Types of Drift Detection Patterns
 - [1] Expand LW & Maintain Existing Prediction Model
 - [2] Shift LW → SW & Maintain Existing Prediction Model
 - [3] Shift LW → SW & **Feature Selection**



Data Management with CD Detection and Adaptation

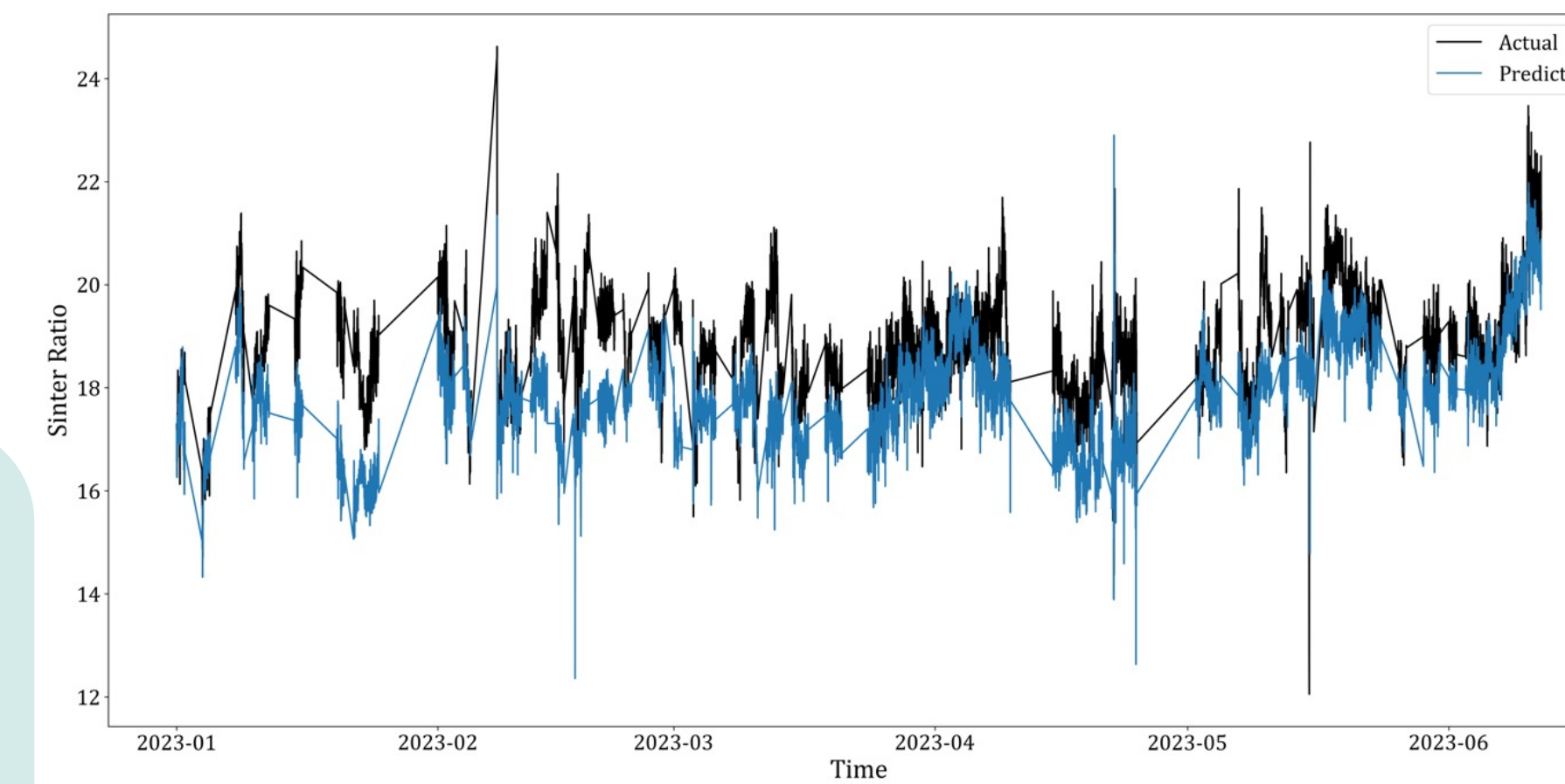
Non-Periodic Model Update for Adaptation to Concept Drift Focused on Buffer and Weights

Key Concept	Explanation
Buffer Period	Refers to the interval during which additional data is collected before updating the model following the detection of CD.
Differential Weighting	A mechanism for adjusting the sensitivity of CD adaptation. Differentiates the influence of data collected during the buffer period.



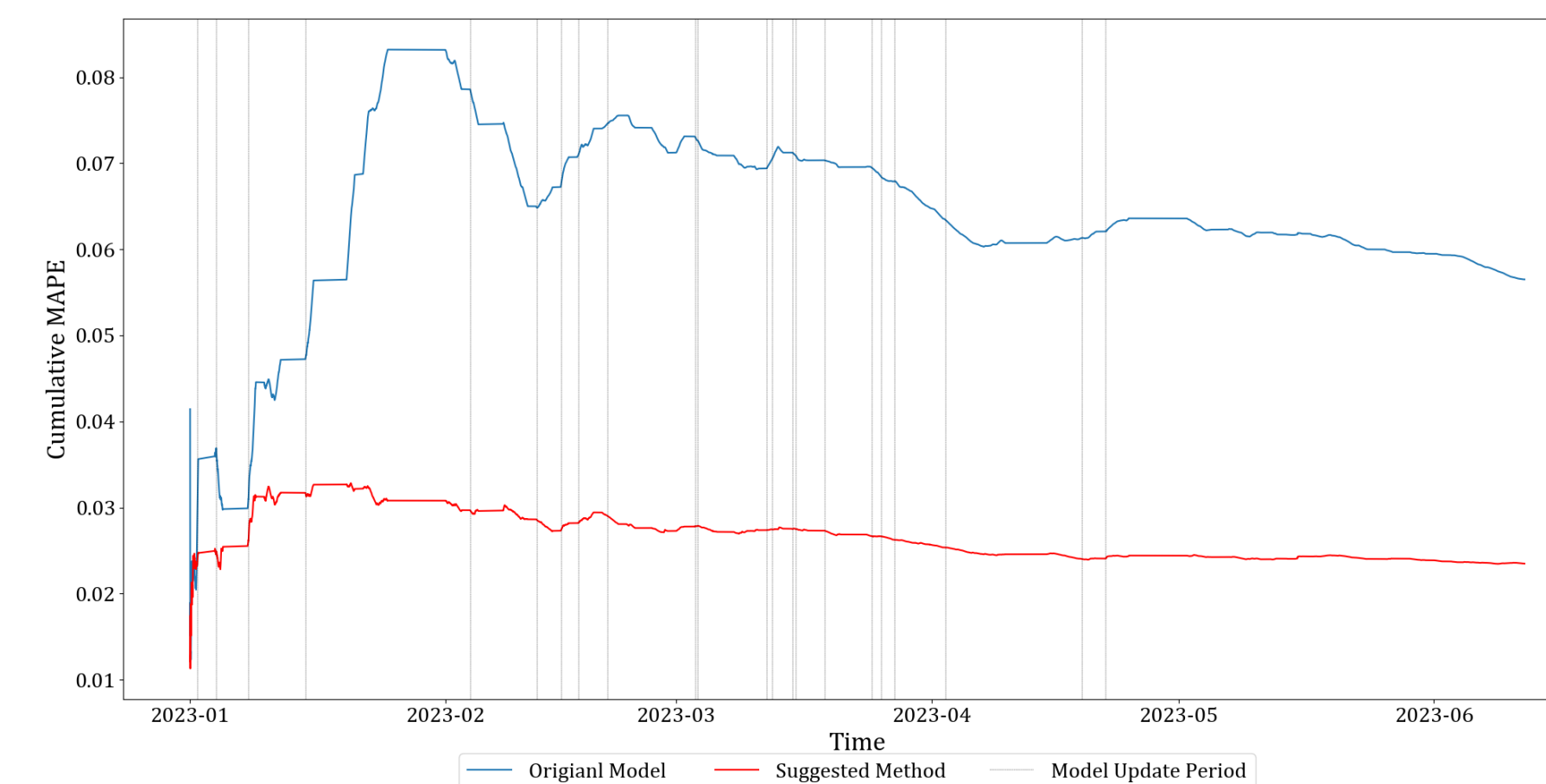
- Weighting functions: Uniform, Stepwise, Linear, and Logarithmic
- Adaptive sensitivity settings for more information are possible.

Case Study – Korean Steel Manufacturing



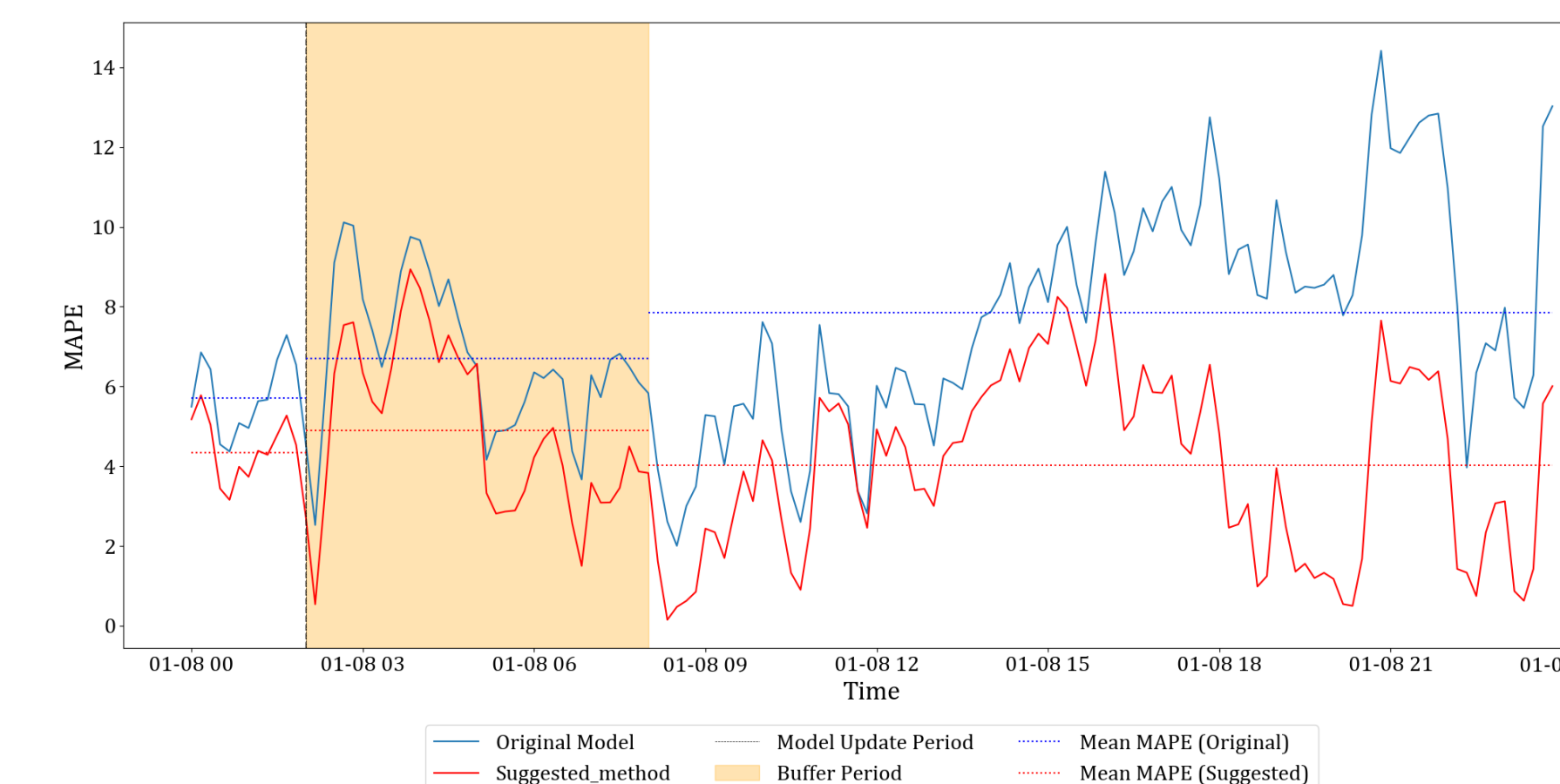
- Model performance **Degradation** over time
- Occurrence of 'Abnormal Interval' caused by **Unstable** Predictions
- Data analysis objective: Predicting the occurrence rate of return fines**

The Result of Application of Model Update Strategy



- The proposed method showed approximately **74% improvement in CTQ** performance compared to the original method.
- The sensitivity of proposed model was reduced while enhancing overall performance.
- The **suggested framework** showed better results across different configurations.

Discussion Points



Continuous Process in Steel Manufacturing

- Ironmaking, steelmaking, casting, rolling processes occur in sequence.
- The sintering process converts iron ore into sinter.
- The quality of the sinter directly affects the final product.**
- Return fines** are produced during sintering.
- Monitoring return fines can predict the quality of the sinter.

Exploratory Data Analysis(EDA)

- Data from the **Korean steel manufacturing industry**.
- Raw data was restructured into dataset consisting of 50,832 rows.
- Feature Discretization Method: MDLP^[3]
- Feature Selection Method: mRMR^[4]

Values (MAPE / CTQ, %)	Update				
	X	Periodic Update	O		
			Buffer Period	Weight	Suggested Framework
Original Model	5.65 / 53.06	3.91 / 72.15	X	X	-
Proposed Model	3.32 / 80.94	2.46 / 88.02	X	Constant	2.35 / 92.39
				Various	2.35 / 92.41
				O	Constant 2.36 / 92.29 Various 2.34 / 92.47

Period	Original Model (MAPE / CTQ, %)	Suggested Framework (MAPE / CTQ, %)
Overall	7.39 / 7.63	4.29 / 40.97
Before Buffer	6.70 / 0	4.88 / 30.76
Buffer	5.70 / 5.40	4.33 / 37.83
After Buffer	7.84 / 9.37	4.03 / 43.75

- The **suggested framework** applying Buffer Period was able to maintain stable performance by controlling the sensitivity to changes in the data environment.

References

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[5] Lim, Dong-Joon, et al. "Development of a virtual metrology system for smart manufacturing: A case study of spandex fiber production." Computers in Industry 145 (2023): 103825.